

Assignment 2: Git and Stream Temperature - Data Download

Grace Bryan

2026-05-07

Data Source

Daily stream temperature data were downloaded from the USGS NWIS database for three sites on the Willamette River from 2010-2023.

Data Download or Load

```
library(dataRetrieval)
library(tidyverse)
library(lubridate)

# Site IDs (USGS format required for API)
siteNumbers <- c("USGS-14211720", "USGS-14211010", "USGS-14166000")

# Toggle: TRUE = download fresh data, FALSE = load saved data

download_data <- TRUE

if(download_data){

  temp_data <- read_waterdata_daily(
    monitoring_location_id = siteNumbers,
    parameter_code = "00010", # Water temp
    statistic_id = "00003",    # Daily mean
    time = c("2010-01-01", "2023-12-31"),
    skipGeometry = TRUE
  )
}
```

```
# Save data locally
write.csv(temp_data, "data/temp_data.csv", row.names = FALSE)

} else {

  temp_data <- read.csv("data/temp_data.csv")
}
```

Data Processing

The raw dataset was cleaned and reshaped for analysis. Relevant columns were selected and renamed for clarity, and time variables (month and year) were extracted to support analysis.

```
# Select and rename relevant columns
temp_data <- temp_data %>%
  select(
    agency_cd = monitoring_location_id,
    site_no = monitoring_location_id,
    Date = time,
    temp_C = value )

# Clean site names
temp_data$site_no <- gsub("USGS-", "", temp_data$site_no)

# Ensure Date format
temp_data$Date <- as.Date(temp_data$Date)

# Extract time variables
temp_data$month <- month(temp_data$Date, label = TRUE)
temp_data$year <- year(temp_data$Date)
```

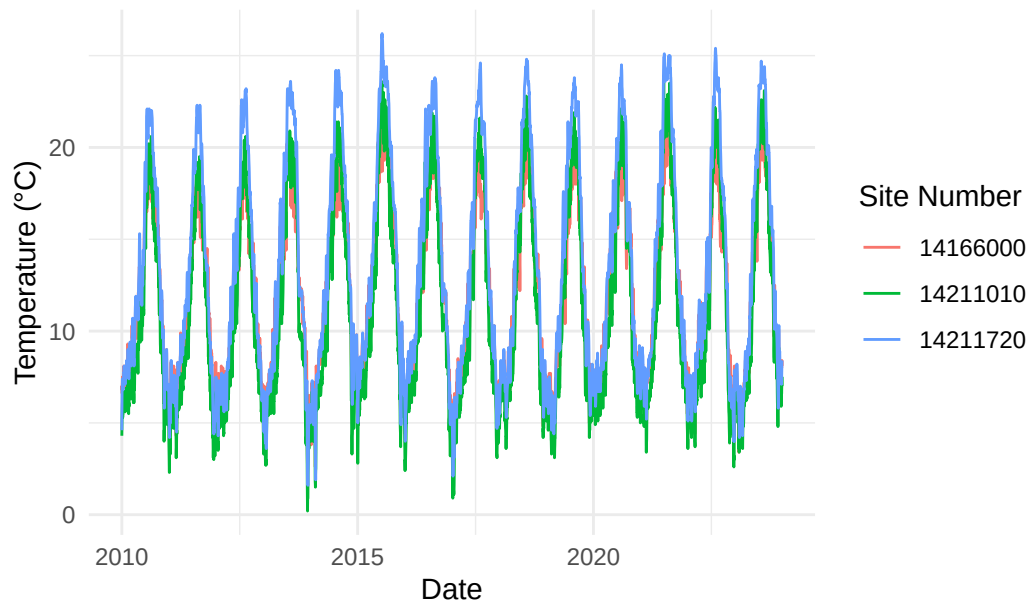
Visualization

The plot below shows temperature variation over time for each monitoring site.

```
# Plot
ggplot(temp_data, aes(x = Date, y = temp_C, color = site_no)) +
  geom_line() +
  labs(
    title = "Stream Temperature at Willamette River Sites",
```

```
x = "Date",
y = "Temperature (°C)",
color = "Site Number" ) +
theme_minimal()
```

Stream Temperature at Willamette River Sites



Summary Statistics

```
temp_data %>%
  group_by(site_no) %>%
  summarise(
    mean_temp = mean(temp_C, na.rm = TRUE),
    max_temp = max(temp_C, na.rm = TRUE),
    min_temp = min(temp_C, na.rm = TRUE)
  )
```

```
# A tibble: 3 x 4
  site_no mean_temp max_temp min_temp
  <chr>      <dbl>    <dbl>    <dbl>
1 14166000    12.0     22.3     2.4
2 14211010    11.2     23.9     0.2
3 14211720    13.4     26.2     1.6
```